

Project 2 Report

EGGG101

Group 135

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Abstract :

The purpose of this study was to compare the inputs and outputs of two similar products: cast iron and non-stick pans. We created an inventory with values of comparable inputs and outputs for the two products. For practical purposes, we focused specifically on the manufacturing and use phases of the cast iron and non-stick pans. The resources we focused on were energy waste, water waste, and aluminum waste. We found that cast iron pans used significantly more water, specifically during the manufacturing process, while the non-stick pans used more energy. Both cast iron and non-stick pans have areas where one is better than the other, but in general, they are equally matched in terms of waste.

Introduction:

In this study we compared two types of pans against each in a life cycle inventory. Using a gate to utilization boundaries we researched the two products, looking at energy consumption, water usage, and aluminum to determine the better product. This study is divided into three main sections: goal and scope, inventory of inputs and outputs, and conclusion and future work. The goal and scope cover the purpose of this survey. The inventory of inputs and outputs covers the data we collected on the resources used by both cast iron and non-stick pans. The conclusion and future work gives a couple closing remarks on our findings and recommendations going forward.

Goal:

The reason for carrying out this study was to figure out if a cast iron pan or a non-stick pan is better for the environment. The intended audience for this study is families who cook with pans, especially cast iron and non-stick pans. The intended use of this report is to give those consumers evidence on why they should choose one pan over the other in order to help out the environment.

Scope:

Based on our intended audience, we decided to measure all data based on kilograms of waste per year of use, assuming an average American family was using the product. The scope is to record and compare the inputs and outputs of the manufacturing and use phases of cast iron and non-stick pans.

Our system diagram includes all life cycle phases of the cast iron and non-stick pans. This includes the raw material acquisition, manufacturing, use, and disposal phases, shown below:

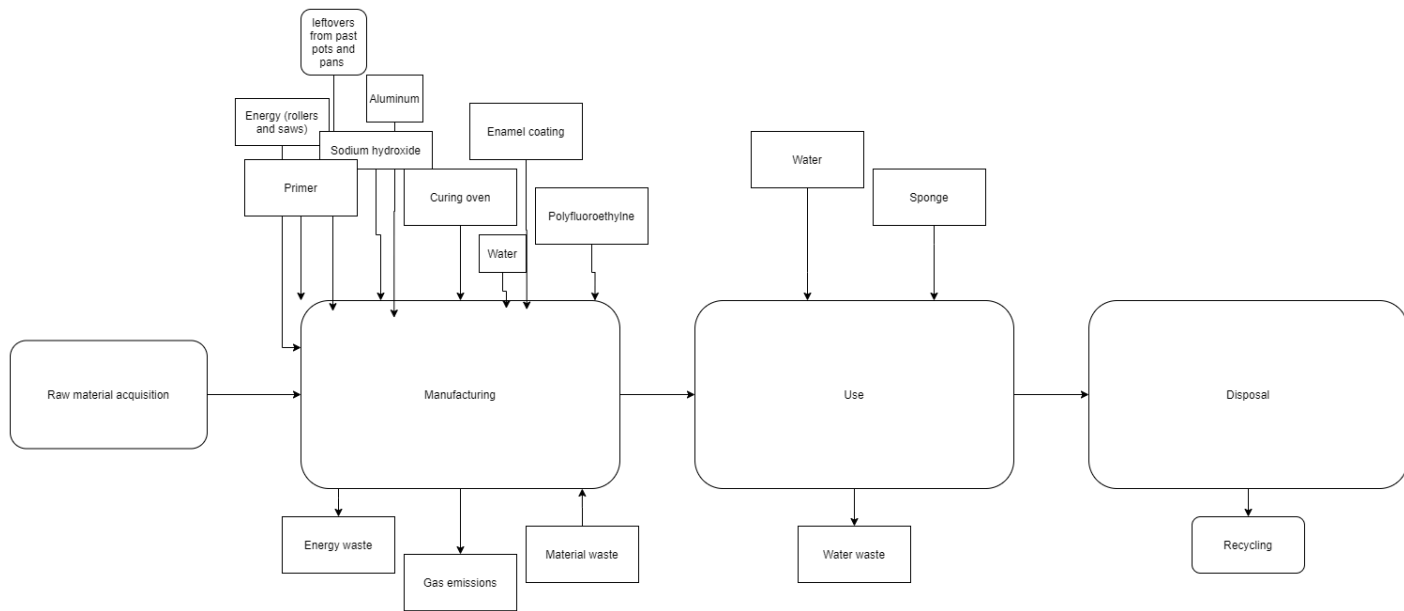


Image 1: System diagram for a non-stick pan

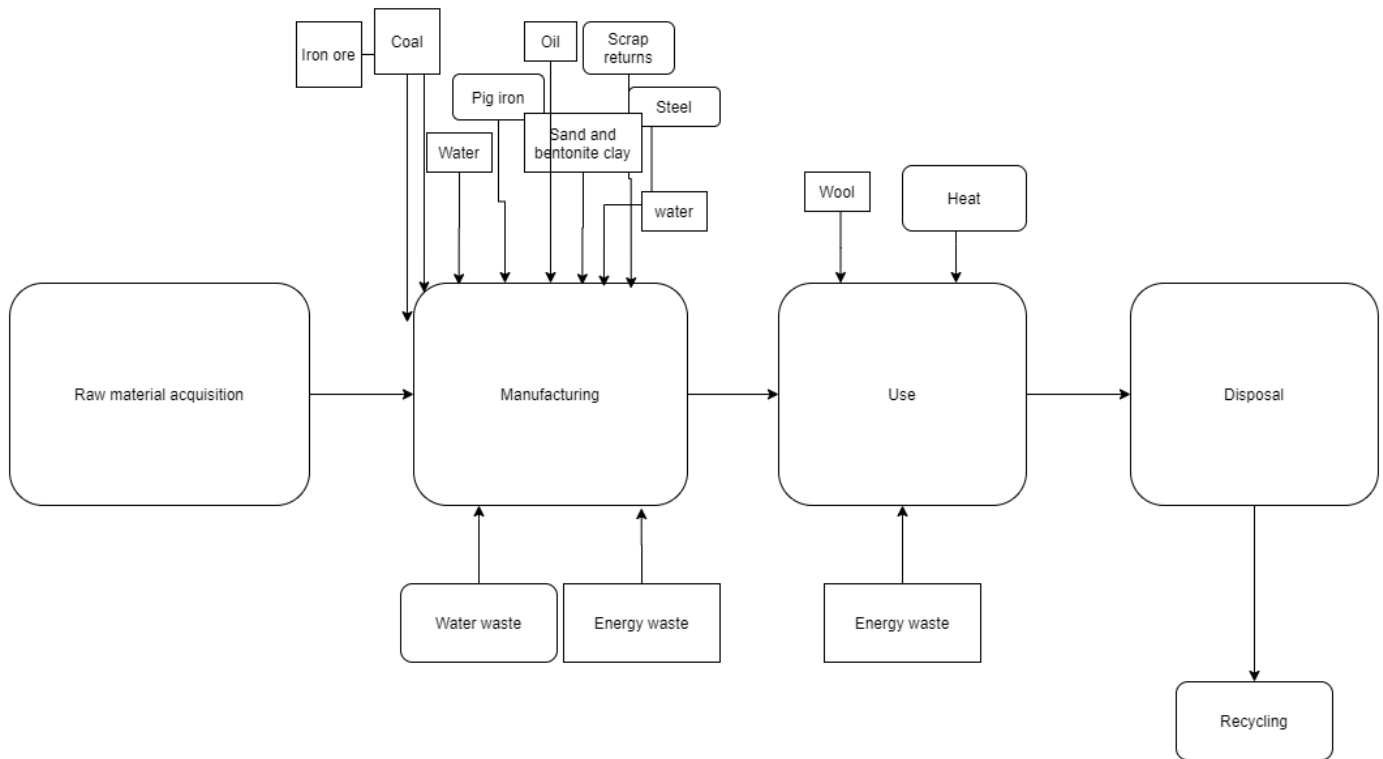


Image 2: System diagram for the cast - iron pan

For the system boundaries, we went from gate to utilization in the life stage of the production process. We decided to cut out the raw material acquisition and the disposal phases of the project, due to the similarities between the inputs and outputs for these phases between cast iron and non-stick pans. We included the manufacturing and use phase of the products along with the inputs and outputs of each of those phases.

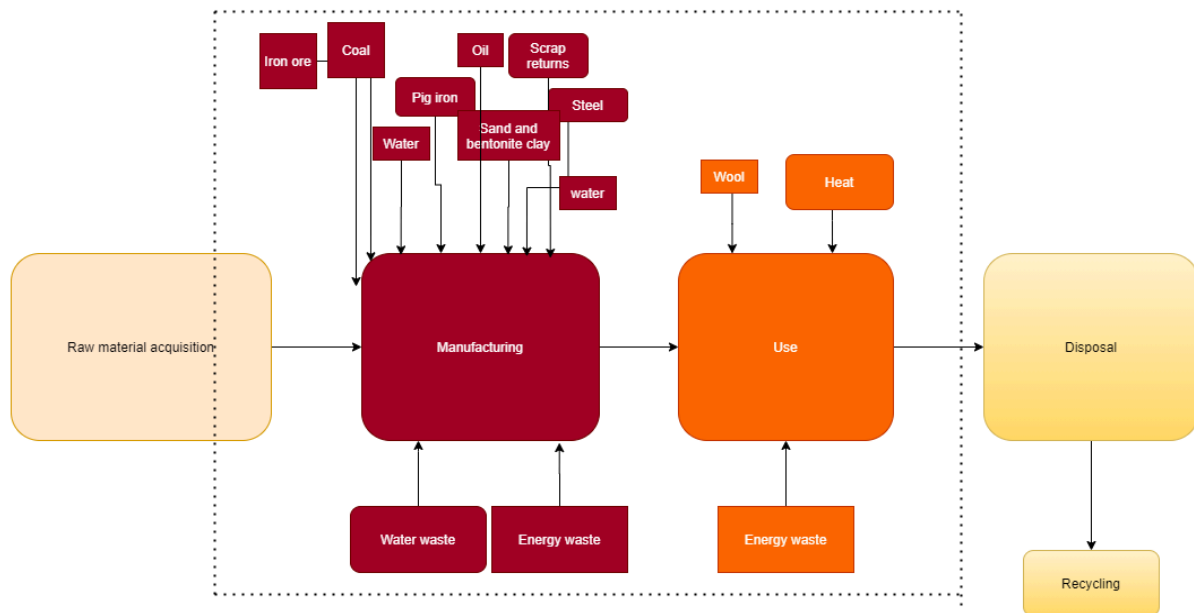


Image 3: System boundary for the cast-iron pan

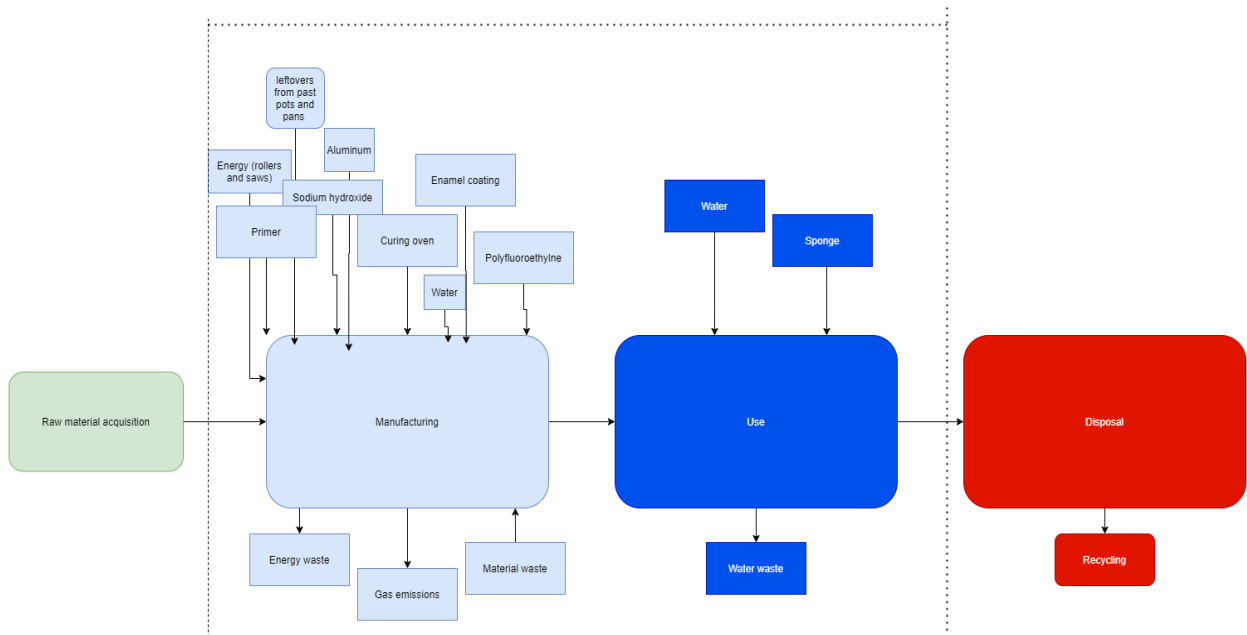


Image 4: System boundary for the non-stick pan

Inventory of Inputs and Outputs :

After researching the products we looked for sources supporting our inputs and outputs to create an inventory comparing the data between the pans. In inventory we compared the inputs of energy, water and aluminum. In our outputs we compared which were also energy, water and aluminum. For energy, we assumed that the source of energy was natural gas, a common fuel source [4]. For more detailed information please visit our appendix.

Table 1 : Inputs of Cast Iron Pan

Life Cycle Stage	Input	Quantity (kg/year of use)
Manufacturing	Aluminum	1/60
	Energy	0.15
	Water	438888
Usage	Energy	0.05
	Water	N/A

Table 2 : Outputs of Cast Iron Pan

Life Cycle Stage	Output	Quantity (kg/year of use)
Manufacturing	Aluminum	N/A
	Energy	0.03
	Water	5795.1
Usage	Energy	N/A
	Water	N/A

Table 3: Inputs of Non-stick pan

Life Cycle Stage	Output	Quantity (kg/year of use)
Manufacturing	Aluminum	1/60
	Energy	0.205
	Water	5000

Usage	Energy	47
	Water	N/A

Table 4: Outputs of Non-stick pan

Life Cycle Stage	Input	Quantity (kg/year of use)
Manufacturing	Aluminum	N/A
	Energy	0.04
	Water	250
Usage	Energy	N/A
	Water	5000

Interpretation/Conclusion:

Based on our data, the amount of materials needed for the production of a cast iron pan and a non-stick pan differ mostly based on energy and water consumption. The processes needed for cast iron formation require far more water because of the need to cast the metal in a mold rather than simply mold a sheet of metal around a punch [9]. The energy needed to create a cast iron pan, however, is smaller than the energy needed to make a non-stick pan.

The vast amount of water needed to create a cast iron pan makes the non-stick pan seem like a better choice. It should be noted that most of the water used in the cast iron manufacturing is recycled [9]. In addition, the non-stick pan creates gas emissions that have negative impacts at high enough levels [14]. So, the cast iron and non-stick pan are evenly matched.

Table 5: Comparison of inputs and outputs for the functional unit

Life Stage	Cast Iron Pan			Non Stick Pan		
Manufacturing	Energy: 6.3 MJ/year of use	Aluminum : 1/60 kg aluminum/ years of use	Water: 438888 kg of water/ year of use	Energy: 8.6 MJ/year of use	Aluminum : 1/60 kg of aluminum/ years of use	Water: 5000 kg/year use

Future Work:

In the future, our study could include more inputs and outputs for the two products. One potentially important byproduct that we did not include in our LCI was gas emissions. In the case

of non-stick pans, the non-stick coating can degrade to create harmful gasses if heated enough [14]. In addition, although we did not have the resources to do so ourselves, obtaining data from a more direct source, such as Lodge factories, could allow us to eliminate possible errors in our data.

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